Geeganage Senuja Dilmith

Electrical Engineering Undergraduate | University of Moratuwa

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in/senuja-dilmith GitHub Portfolio

Summary

Dedicated Electrical Engineering undergraduate at the University of Moratuwa with a strong passion for IoT, machine learning, electronics, and robotics. Backed by extensive hands on experience in building advanced hardware,software systems, custom PCB fabrication, and actively participating in prominent innovation and IoT competitions.

Educational Qualifications

BSc. Eng. (Hons.) in Electrical Engineering 2024 – Present
University of Moratuwa, Sri Lanka | Cumulative GPA: 3.76/4.0 (upto Semester 04)

Secondary & School Education 2009 – 2023
Richmond College (2020–2023) & All Saints' College (2009–2020)

- 3 A's in G.C.E. Advanced Level (A/L) Examination 2022 (2023)- Physical Science Stream
- 9 A's in G.C.E. Ordinary Level (O/L) Examination 2019

Achievements

1st Runners-Up | SLIoT Challenges 2026 by University of Moratuwa
Top 5 Finalist | Biofusion 2025 by University of Jayewardenepura
Finalist | Brainstorm 2026 by APIIT Campus
Finalist | The Pitch 60 Competition by University of Moratuwa
1st Runners-Up | Cosmotron Rover Competition by University of Ruhuna

Projects & Products

GTricks Pathfinder | ***Offline IoT Safety & Navigation Post System***

- Built a private LoRa backbone network featuring custom dual-layer printed circuit boards and a local offline web dashboard interface to provide digital trail maps and SOS transmission in cellular dead zones. GitHub | LinkedIn

The Phenomena Detector | ***Chest X-Ray Pneumonia Detection Software***

- Built a software solution powered by a custom strong machine learning module developed by team GMora, capable of scanning hard or soft copies of chest X-rays to detect pneumonia. GitHub

Smart Baby Care System | ***Measurements and Instruments Module Project***

- Designed an advanced IoT and LabVIEW monitoring solution integrating an ESP32-CAM hosted via Hugging Face machine learning model for baby and crying detection, automated climate control fans, automated mosquito nets, and custom-calibrated potentiometer tilt-measurement safety mechanisms. GitHub | LinkedIn

The Green Dot | ***IoT Freshness-Monitoring Device***

- Developed a compact, low-cost IoT freshness-monitoring device featuring multi-sensor fusion (MQ-135 gas sensors for VOCs/ethylene/ammonia, BME280 for micro-climate, and vibration sensors) integrated with custom industrial-grade printed circuit boards and a robust ML-based equation to accurately sort the food queue and prioritize inventory freshness. LinkedIn

Rover Simulation Project | ***Webots Autonomous Rover***

- Design and program a fully autonomous rover within the Webots simulation environment, utilizing 360-degree camera arrays, GPS localization, and kinematic path-planning algorithms to traverse unstructured terrain and locate target objectives. GitHub

Automatic Brightness Control LED System | ***EE3204 Research Project***

- Researched and designed an adaptive illuminance control system in MATLAB/Simulink that dynamically balances lighting between 300 and 500 Lx based on occupancy detection and ambient monitoring, achieving a 75.97% reduction in energy consumption. LinkedIn

Advanced Light Intensity Indicator | ***Semester 3 Digital Signal Processing Module Project***

- Designed and built a hardware-based signal processing system using Logic ICs, Flip-flops, and hardware noise filtration to calculate long-term moving averages and stability control for raw light-dependent resistor sensor data. LinkedIn

FixitNow | ***Modular Software Development Module Project***

- Engineered the backend architecture for an on-demand home repair and maintenance marketplace using Java 21, Spring Boot 4, and MySQL 8 database persistence via Spring Data JPA, deployed on AWS EC2 behind an Nginx reverse proxy with CI/CD automation. GitHub

Industrial Experiences

- Conducted technical site visits to major thermal power generation facilities, including the Yugadanavi Combined Cycle Power Plant (300MW), Sobadhanavi Combined Cycle Power Plant (350MW), and Kelanitissa Power Plant, observing large scale turbine operations, generator synchronization, and combined cycle efficiency.

Certifications

- **Supervised Machine Learning - Regression and Classification :** Coursera certification completed under professional machine learning frameworks.

Leadership & Extra-Curricular Activities

Committee Leader (Company Calling Lead) | ***CareerX (2026)***

- Spearheaded company outreach and relationship management to build a direct recruitment platform connecting students with corporate partners.

Committee Leader (Design Lead) | ***Prakampana (2026)***

- Managed visual branding, creative assets, and design execution for university cultural events.

Committee Leader (Design Lead) | ***RTC and En'viera Events (2026)***

- Directed design operations for the Round Table Conference (RTC) and En'viera (Department of Electrical Engineering alumni gathering).

Member & Design Committee Leader | ***Robotics and Automation Society (RAS), University of Moratuwa***

- Led design initiatives and creative campaigns for RAS events, including **Beyond the Pages** and **MazeX** (an inter-university micromouse maze competition).

Student Coordinator | ***EESpire Career Fair (2025 and 2026)***

- Coordinated the Department of Electrical Engineering's annual career fair, bridging final-year engineering students with industry partners for employment and internships.

Member of Marketing Team | ***EESoc***

- Active participant in departmental electrical engineering society activities and programs.

Volunteer & Motivational Speaker | ***Sasnaka Sansada, Galle District (2023 – Present)***

- Conducted motivational seminars and educational guidance sessions for youth and students.

Former Mathematics Trainer | ***Southern Province Math Olympiad Pool (2023–2024)***

- Designed advanced training curricula and tutored high-achieving secondary students in competitive mathematics.

Volunteer | ***Soyuru Sathkara Program by UoM (2025) & Shilpasara Program by Leo Club of UoM (2024)***

- Visited schools to teach mathematics to Ordinary Level (O/L) students, delivered motivational sessions, and conducted community educational outreach.

Vice Captain | ***Debate Club, All Saints' College (2019)***

- Directed debate preparation, public speaking, and inter-school competitions.

Skills

- **Hardware and IoT Tools**: ESP32, STM32, LoRa modules, NI DAQ card, Arduino, PCB design tools
- **Software Frameworks and Tools**: YOLO, SolidWorks, Figma, MATLAB, LabVIEW, Webots, HuggingFace, Git/GitHub, PSCAD, PSSE
- **Programming Languages**: Python, C, C++, Java, JavaScript, MATLAB, HTML
- **Soft Skills**: Public and motivational speaking, graphic design, script writing, team management, event planning, Teaching

References

Prof. Buddhika Jayasekara
Head of the Department
Department of Electrical Engineering
University of Moratuwa

Dr. V. Logeeshan
Senior Lecturer
Department of Electrical Engineering
University of Moratuwa